



AMERICAN INTERNATIONAL UNIVERSITY-BANGLADESH (AIUB)

Dept. of Computer Science

Faculty of Science and Technology

CSC2210: OBJECT ORIENTED PROGRAMMING 2

Spring 2025-2026

Section: EE

Group No: 00

Project Report On

Gadget Shop Management System

Supervised By

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Submitted By:

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Obtained Marks for CO2 and CO3 (Description given in the following page)					
Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)
Evaluation Criteria (CO2)	Total =		Evaluation Criteria (CO3)		Total =
Requirement fulfillment			Organization of the application		
Validation			Representation and Integration of Database		
Verification			Graphical User Interface		

CO2: Display and verify the mean of a real-life Project using the concepts of C# Graphical User Interface based environment with database integration to depict a desktop-based application.

Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)
Evaluation Criteria	Evaluation Definition				
Requirement fulfillment	Fails to demonstrate any understanding of real-life scenario-based project development or functional requirement identification. There is no attempt to depict a project or identify functional requirements accurately.	Demonstrates limited understanding of real-life scenario-based project development and functional requirement identification. The project depicted lacks coherence or relevance to real-life scenarios, and functional requirements are inaccurately identified or insufficiently described.	Presents a basic depiction of a real-life scenario-based project and identifies some functional requirements. However, the project lacks depth or complexity, and some functional requirements may be vaguely defined or missing key details.	Effectively demonstrates a realistic scenario-based project and accurately identifies most functional requirements. The project is well-developed with appropriate complexity, and functional requirements are clearly articulated with relevant details.	Exhibits an exceptional understanding of real-life scenario-based project development and accurately identifies all functional requirements. The project is meticulously developed with thorough attention to detail, reflecting a comprehensive understanding of Object-Oriented Programming project development activities.
Validation	Fails to demonstrate any understanding or	Demonstrates limited understanding of validation forms and data	Shows a basic understanding of validation forms and data validation	Effectively demonstrates the use of validation forms and	Exhibits an exceptional understanding and implementation

	implementation of validation forms in their system. There is no attempt to deal with data validation, and validation requirements are completely ignored or incorrectly applied.	validation techniques. While some attempt may be made to implement validation, it is incomplete or poorly executed, leading to inadequate handling of data validation.	techniques. They attempt to implement validation, but some aspects may be missing or incorrectly implemented, resulting in partial or inconsistent handling of data validation.	implements data validation techniques. Validation is mostly accurate and comprehensive, ensuring the proper handling of data input and verification in the system.	of validation forms and data validation techniques. Validation is meticulously implemented with thorough attention to detail, ensuring robust data validation procedures and contributing to the overall reliability and integrity of the system.
Verification	Fails to demonstrate any attempt to verify the system data or functional requirements. There is no evidence of understanding or implementation of verification processes, and data flow is not considered.	Demonstrates limited understanding of verification processes and data flow in the system. Verification attempts are incomplete or inaccurate, and there is insufficient consideration given to ensuring data integrity and functionality.	Shows a basic understanding of verification processes and attempts to verify system data. However, verification efforts may be inconsistent or lack thoroughness, and there may be gaps in ensuring proper functional requirements	Identifies and verifies system data, ensuring proper functional requirements are met. Verification efforts are mostly accurate and thorough, with attention to ensuring data integrity and appropriate data flow within the	Exhibits an exceptional understanding of verification processes and meticulously verifies system data. Verification efforts are comprehensive and precise, with a keen focus on ensuring all functional requirements are met and maintaining

			and data flow.	system.	proper data flow throughout the system.
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CO3: Prepare and Explain a real-life desktop based application synthesizing several component of C# along with development tools to adhere the given requirements.

Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)
Evaluation Criteria	Evaluation Definition				
Organization of the application	Fails to identify any suitable real time application or requirements for project development activities related to OOP.	Limited understanding about the project scopes and scenarios or identification of functional requirements.	Lacks depth or relevance to OOP project development activities and may contain inaccuracies. Real-life scenarios are mentioned, but the discussion lacks depth or clarity.	Consider and integrate the idea of several core aspects of the project along with relevance to real-life scenarios. Demonstrating a solid understanding of the application presentation.	Generalize and exhibits an exceptional understanding of project preparation according to a to real-life scenarios. Also contains proper and insightful identification of the system which is comprehensive and precise.
Representation and Integration of Database	Fails to identify and present any understanding or implementation	Limited understanding of the database concepts or their proper way of using in	Lacks depth or relevance to database integration with the application.	Integrate the database with the forms properly and implements it with proper	Exhibits an exceptional understanding and implementation of database

	of database. Also failed to integrate the data with the project itself.	a real time project. While some attempt may be made to implement but it is incomplete or poorly executed, leading to inadequate design.	Shows a basic understanding but some aspects may be missing or incorrectly implemented, resulting in partial or inconsistency. May lack proper normalization.	validation which is mostly accurate and comprehensive, ensuring the proper handling of data input and verification along with general normalization.	ensuring attention to detail, and robust data manipulation procedures and contributing to the overall clarity.
Graphical User Interface	Fails to present or prepare GUI based application interfaces. There is no evidence of creating or integrating such things according to their usefulness.	Limited understanding of graphical user interfaces. Lack of design knowledge. Very poor attempt to make such things which are currently obsolete or can't be identified as coherent.	Shows a basic understanding of creating user interfaces. Most of them are interconnected but maybe some of them lack it. However, most of it can be described as user friendly.	Effectively identifies and meet the consider the simplicity. Design related works are mostly accurate and taken proper attention to ensuring a user-friendly coherent system.	Exhibits an exceptional work design following a high standard of simple and elegant work. Several controls and mechanism has been organized in a preferred way according to the coherent usage .

The Electric Gadget Management System is a combination of a desktop-based and web-based application that will help simplify the process of running an electronic gadget shop. It creates an effortless interaction between customers and the system admin. The customers have the ability to search for a broad list of gadgets available (smart phones, laptops, smart televisions etc.) categorize them, add products to their cart and check out using the payment gateway. The administrator has a strong dashboard where they can manage the product stock (performing CRUD operations), real-life statistics (total revenue and orders made) and validate the payments made by customers.

The system has been developed using C# .NET, and employs a layered architecture, incorporating SQL server for efficient management of the database. The GUI has been created with a contemporary look, emphasizing aesthetics and usability.

Chapter 2: User Stories

This chapter explains the main goals and interactions of different users in the **Electric Gadget Management System**.

2.1 Customer User Stories

When people use this system, they can look around, search for things, and buy electronic gadgets. They can check out products by what kind of thing they are, look at pictures and details, compare them to other products, and put them in their shopping cart. To pay for the things they want, they can use things like bKash, a Credit Card, or they can pay when their stuff is delivered, which is called Cash on Delivery. After they've bought something, they can tell others what they thought of it by giving it a rating and writing a review, which helps other people who are thinking of buying the same thing. This way, people can share their experiences and help each other make good choices when they're shopping.

2.2 Admin User Stories

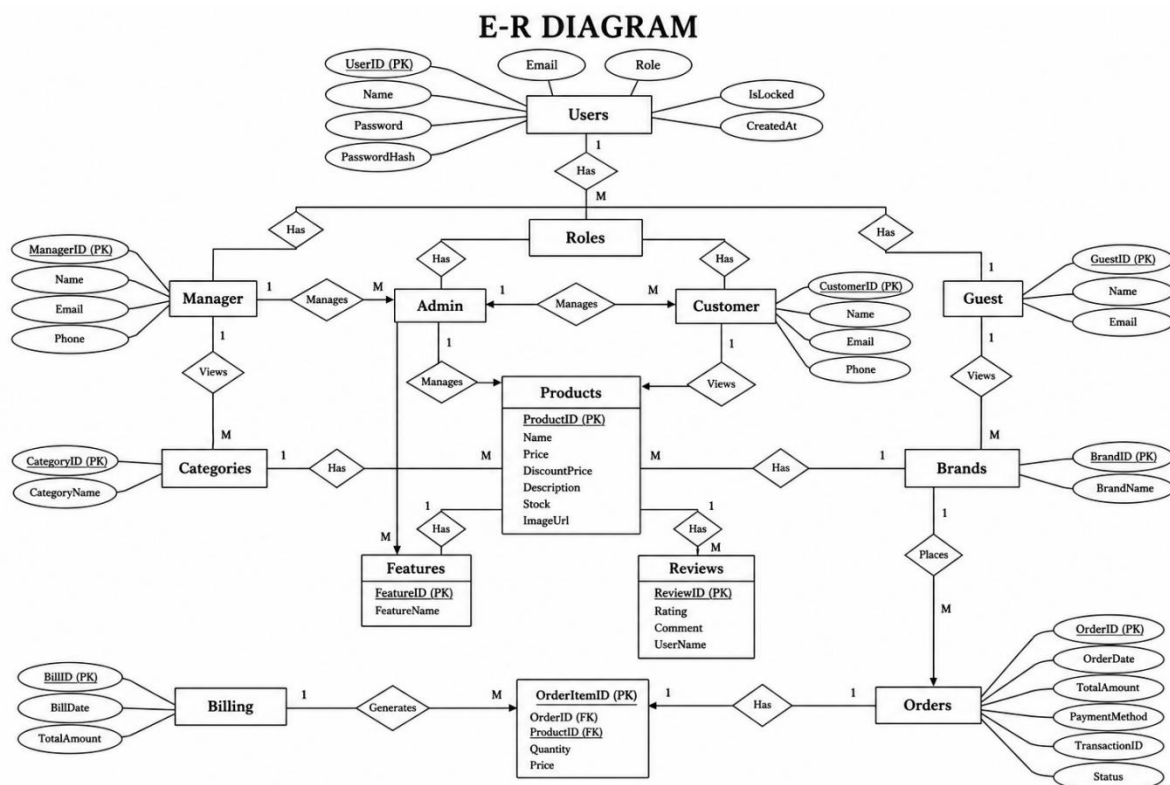
Admins manage the store's products and inventory. They can add, update, or remove products and organize them into categories and brands. Admins monitor stock levels, manage customer

orders, and update order statuses such as Pending, Paid, or Shipped. They also view sales reports and manage discounts or promotional offers to improve business performance.

2.3 Super Admin User Stories

Super Admins manage the entire system and ensure security, scalability, and smooth operation. They oversee branches, manage user accounts, and monitor audit logs for transparency. Super Admins also handle system backups, global settings like currency and timezone, and SaaS subscription management to maintain uninterrupted service.

Chapter 3: System Functionalities



1. Users: UserID, Name, Password, PasswordHash, Role, Email, IsLocked, CreatedAt
2. Categories: Id, Name
3. Brands: Id, Name, CategoryId
4. Products: Id, Name, Price, DiscountPrice, Description, Stock, ImageUrl, BrandId, CategoryId
5. Features: Id, FeatureName, ProductId
6. Reviews: Id, Rating, Comment, UserName, ProductId
7. Orders: Id, CustomerName, CustomerEmail, OrderDate, TotalAmount, PaymentMethod, TransactionId, Status

8. OrderItems: Id, OrderId, ProductId, Quantity, Price

DATABASE TABLE STRUCTURE

1. Users

Column Name	Data Type	Constraint
UserID	NVARCHAR(50)	Primary Key
Name	NVARCHAR(100)	NOT NULL
Password	NVARCHAR(100)	NOT NULL
PasswordHash	NVARCHAR(MAX)	NULL
Role	NVARCHAR(50)	NOT NULL
Email	NVARCHAR(100)	UNIQUE
IsLocked	BIT	NOT NULL DEFAULT 0
CreatedAt	DATETIME2	NOT NULL DEFAULT GETDATE()

-----2.Categories

Column Name	Data Type	Constraint
Id	INT	Primary Key, IDENTITY(1,1)
Name	NVARCHAR(100)	NOT NULL

-----3.Brands

Column Name	Data Type	Constraint
Id	INT	Primary Key, IDENTITY(1,1)

Name	NVARCHAR(100)	NOT NULL
CategoryId	INT	Foreign Key → Categories(Id), ON DELETE CASCADE

-----4.Products

Column Name	Data Type	Constraint
Id	INT	Primary Key, IDENTITY(1,1)
Name	NVARCHAR(255)	NOT NULL
Price	DECIMAL(18,2)	NOT NULL
DiscountPrice	DECIMAL(18,2)	NULL
Description	NVARCHAR(MAX)	NULL
Stock	INT	NOT NULL DEFAULT 0
ImageUrl	NVARCHAR(MAX)	NULL
BrandId	INT	Foreign Key → Brands(Id)
CategoryId	INT	Foreign Key → Categories(Id)

-----5.Features

Column Name	Data Type	Constraint
Id	INT	Primary Key, IDENTITY(1,1)
FeatureName	NVARCHAR(200)	NOT NULL
ProductId	INT	Foreign Key → Products(Id), ON DELETE CASCADE

-----6.Reviews

Column Name	Data Type	Constraint
-------------	-----------	------------

Id	INT	Primary Key, IDENTITY(1,1)
Rating	INT	NOT NULL
Comment	NVARCHAR(MAX)	NULL
UserName	NVARCHAR(100)	NULL
ProductId	INT	Foreign Key → Products(Id), ON DELETE CASCADE

-----7.Orders

Column Name	Data Type	Constraint
Id	INT	Primary Key, IDENTITY(1,1)
CustomerName	NVARCHAR(100)	NOT NULL
CustomerEmail	NVARCHAR(100)	NULL
OrderDate	DATETIME2	NOT NULL DEFAULT GETDATE()
TotalAmount	DECIMAL(18,2)	NOT NULL
PaymentMethod	NVARCHAR(50)	NULL
TransactionId	NVARCHAR(100)	NULL
Status	NVARCHAR(50)	NOT NULL DEFAULT 'Pending'

8. OrderItems

Column Name	Data Type	Constraint
Id	INT	Primary Key, IDENTITY(1,1)
OrderId	INT	Foreign Key → Orders(Id), ON DELETE CASCADE
ProductId	INT	Foreign Key → Products(Id), ON DELETE CASCADE

Quantity	INT	NOT NULL
Price	DECIMAL(18,2)	NOT NULL

➤ NOTES / RELATIONSHIPS

1. A Category can have multiple Brands and Products.
2. A Brand can have multiple Products.
3. A Product can have multiple Features and Reviews.
4. An Order can have multiple OrderItems.
5. Each OrderItem belongs to one Product.
6. Email in Users table is unique.
7. Features and Reviews are automatically deleted when related Product is deleted.
8. OrderItems are automatically deleted when related Order is deleted.

```
-- =====  
-- 1. User Registration & Authentication  
-- =====
```

-- Sign Up (Registration)

```
INSERT INTO Users (UserID, Name, Password, Email, Role, CreatedAt, IsActive, IsLocked)  
VALUES ('C-101', 'John Doe', 'password123', 'john@example.com', 'Customer', GETDATE(), 1, 0);
```

-- Sign In (Login)

```
SELECT UserID, Name, Role, BranchId  
FROM Users  
WHERE Email = 'john@example.com'  
AND Password = 'password123'  
AND IsActive = 1;
```

-- Update Password

```
UPDATE Users  
SET Password = 'newSecurePassword789'  
WHERE UserID = 'C-101';
```

-- Lock User (After Failed Attempts)

```
UPDATE Users  
SET IsLocked = 1,  
    FailedAttempts = FailedAttempts + 1  
WHERE Email = 'john@example.com';
```

```
-- =====  
-- 2. Product & Inventory Management  
-- =====
```

```
-- Add New Product
```

```
INSERT INTO Products (Name, Price, Description, Stock, CategoryId, BrandId, ImageUrl,  
IsPublished)  
VALUES ('iPhone 15 Pro', 120000.00, 'Titanium design, A17 Pro chip', 50, 1, 1,  
'/images/iphone15.jpg', 1);
```

```
-- Update Stock Level
```

```
UPDATE Products  
SET Stock = Stock - 1  
WHERE Id = 10;
```

```
-- Search Products by Category
```

```
SELECT p.*, c.Name AS CategoryName  
FROM Products p  
JOIN Categories c ON p.CategoryId = c.Id  
WHERE c.Name = 'Smartphones'  
AND p.IsPublished = 1;
```

```
-- Apply Seasonal Discount
```

```
UPDATE Products  
SET DiscountPrice = Price * 0.9  
WHERE CategoryId = 2;
```

```
-- =====  
-- 3. Sales & Order Management  
-- =====
```

```
-- Create New Order
```

```
INSERT INTO Orders (CustomerName, CustomerEmail, OrderDate, TotalAmount, Status,  
PaymentMethod)  
VALUES ('John Doe', 'john@example.com', GETDATE(), 108000.00, 'Pending', 'Credit  
Card');
```

```
-- Add Items to Order
```

```
INSERT INTO OrderItems (OrderId, ProductId, Quantity, Price)
```

```
VALUES (501, 10, 1, 108000.00);
```

```
-- Track Order Status
```

```
SELECT Status, OrderDate, TotalAmount
```

```
FROM Orders
```

```
WHERE Id = 501;
```

```
-- Update Payment Status
```

```
UPDATE Orders
```

```
SET Status = 'Paid',
```

```
    TransactionId = 'TXN778899'
```

```
WHERE Id = 501;
```

```
-- =====  
-- 4. Branch & Staff Management  
-- =====
```

```
-- Assign Manager to Branch
```

```
UPDATE Branches
```

```
SET ManagerId = 'A-005'
```

```
WHERE Id = 1;
```

```
-- List All Active Branches
```

```
SELECT Name, Location, CreatedAt
```

```
FROM Branches
```

```
WHERE IsActive = 1;
```

```
-- Get Employees by Branch
```

```
SELECT Name, Role, Email
```

```
FROM Users
```

```
WHERE BranchId = 1;
```

```
-- =====  
-- 5. Customer Feedback & Reviews  
-- =====
```

```
-- Post a Review
```

```
INSERT INTO Reviews (ProductId, UserName, Rating, Comment, CreatedAt)
```

```
VALUES (10, 'John Doe', 5, 'Amazing performance and camera!', GETDATE());
```

```

-- Calculate Product Rating
SELECT ProductId,
       AVG(CAST(Rating AS DECIMAL)) AS AverageRating,
       COUNT(*) AS TotalReviews
FROM Reviews
WHERE ProductId = 10
GROUP BY ProductId;

-- =====
-- 6. System Administration & Audit
-- =====

-- Log System Action (Audit Trail)
INSERT INTO AuditLogs (UserId, UserName, Action, Module, Details, Timestamp)
VALUES ('A-001', 'Admin_User', 'DELETE', 'Product', 'Deleted Product ID: 45', GETDATE());

-- View Recent System Activity
SELECT TOP 50 *
FROM AuditLogs
ORDER BY Timestamp DESC;

-- Fetch System Settings
SELECT CompanyName, Currency, SupportEmail
FROM SystemSettings
WHERE Id = 1;

```

Chapter 4: Screenshot of forms:

⚡ Electric Gadget

Smart Gadgets, Better Living

Quality products. Trusted service.
Better life.

Don't have an account? [Sign up](#)

Welcome back

Login to continue

Email address

Password

[Forgot password?](#)

Login

⚡ Electric Gadget

Smart Gadgets, Better Living

Create an account and explore
the latest electric gadgets.

Already have an account? [Log in](#)

Create your account

Sign up to get started

Full Name

Email Address

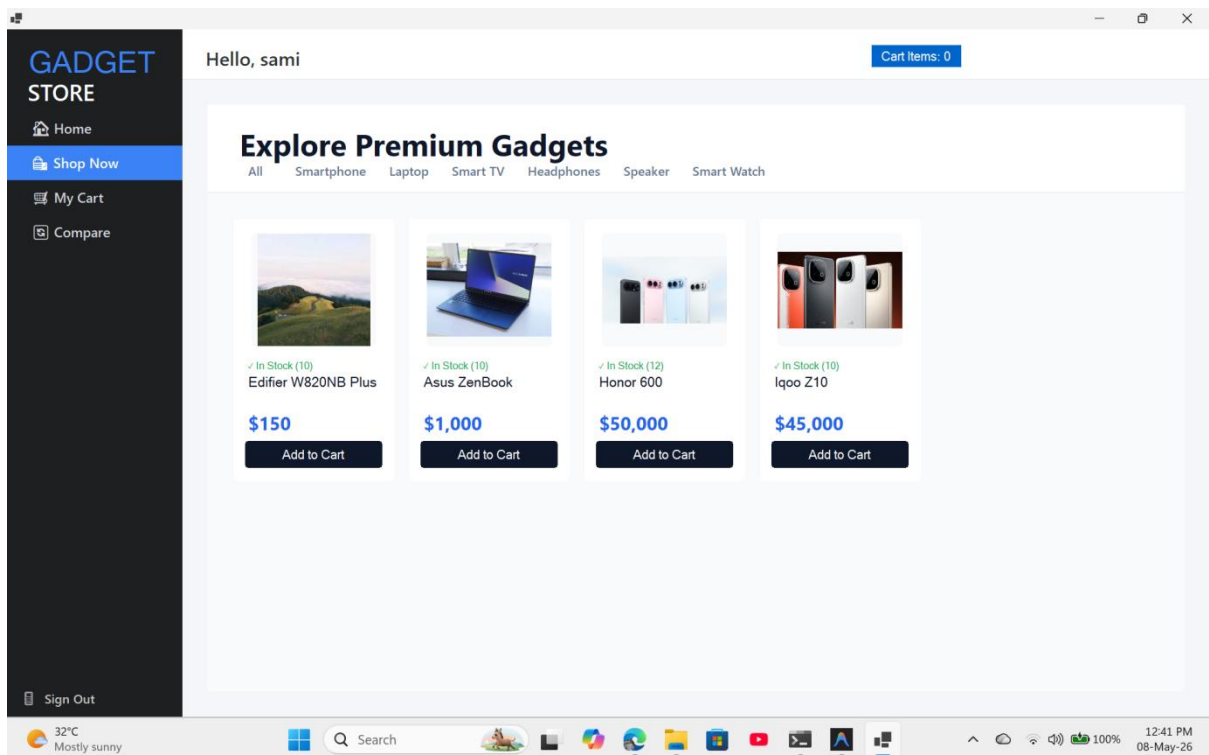
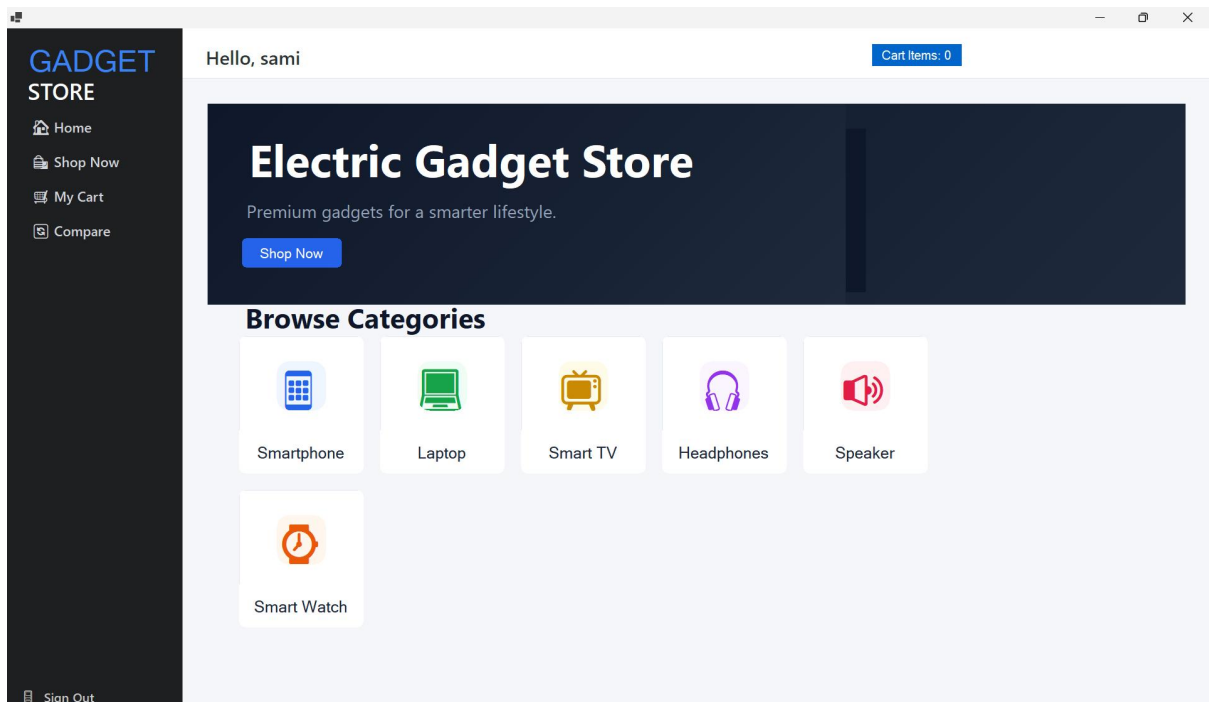
Phone Number (Optional)

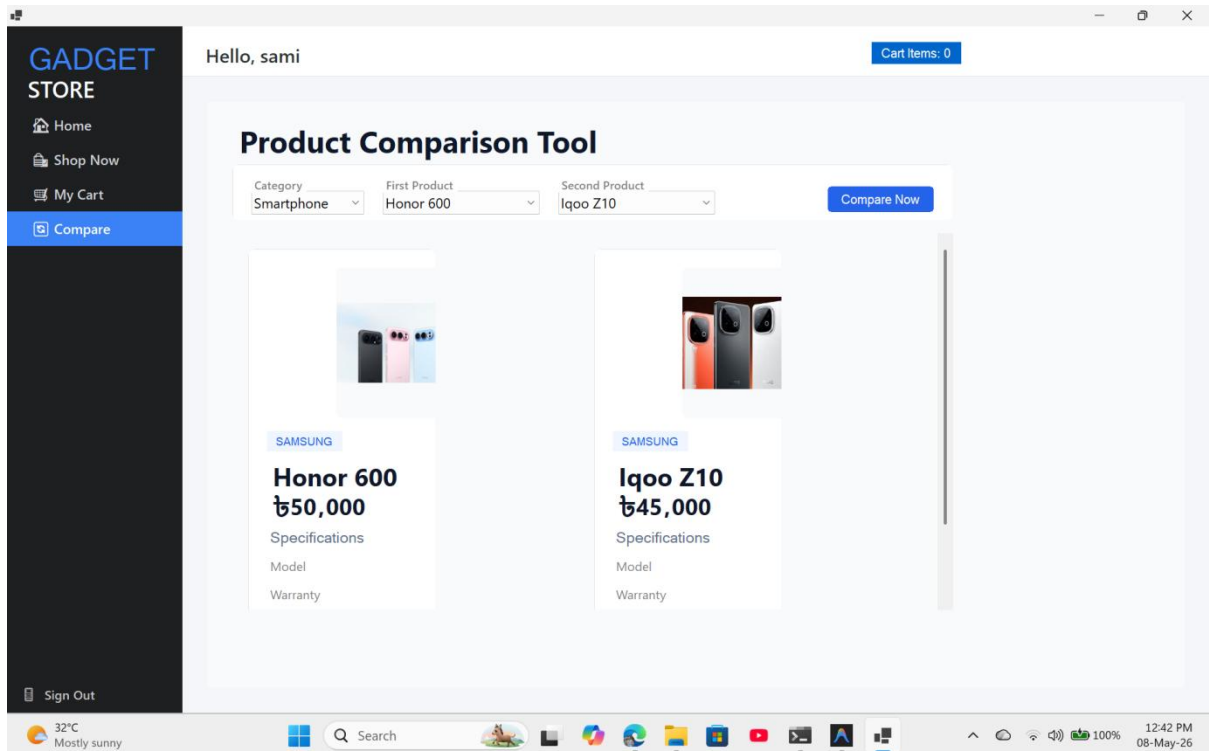
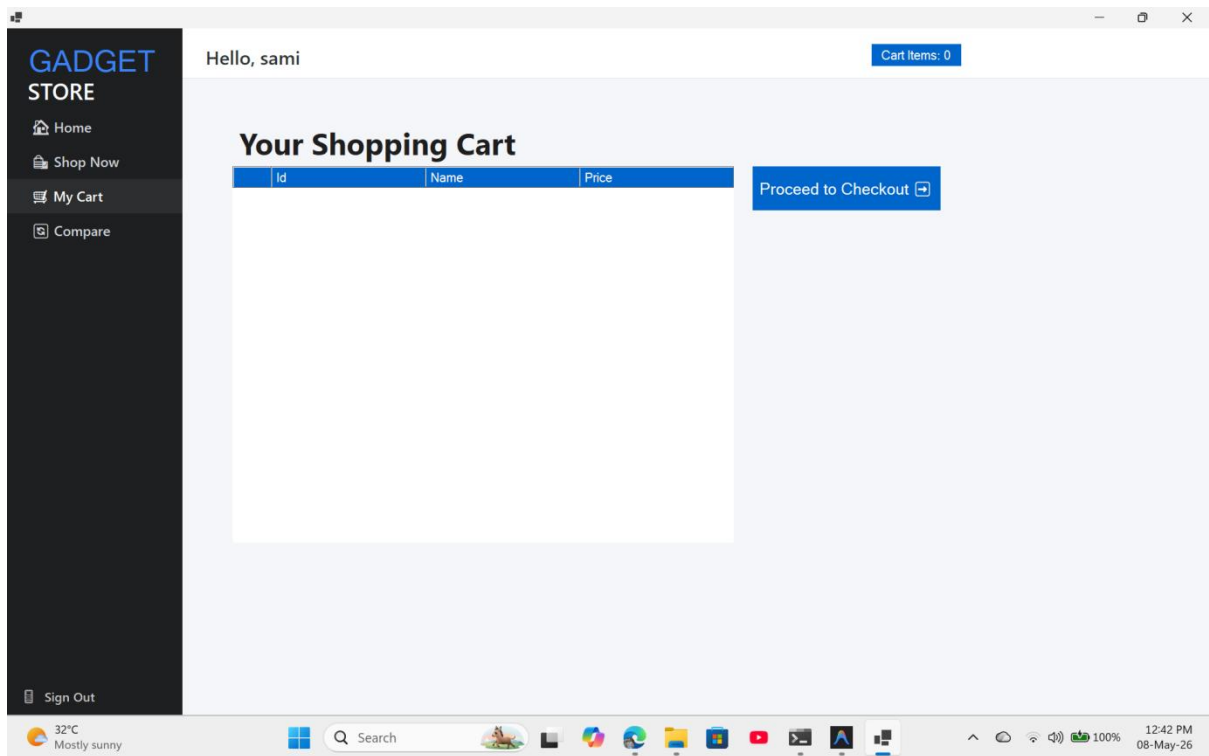
Password

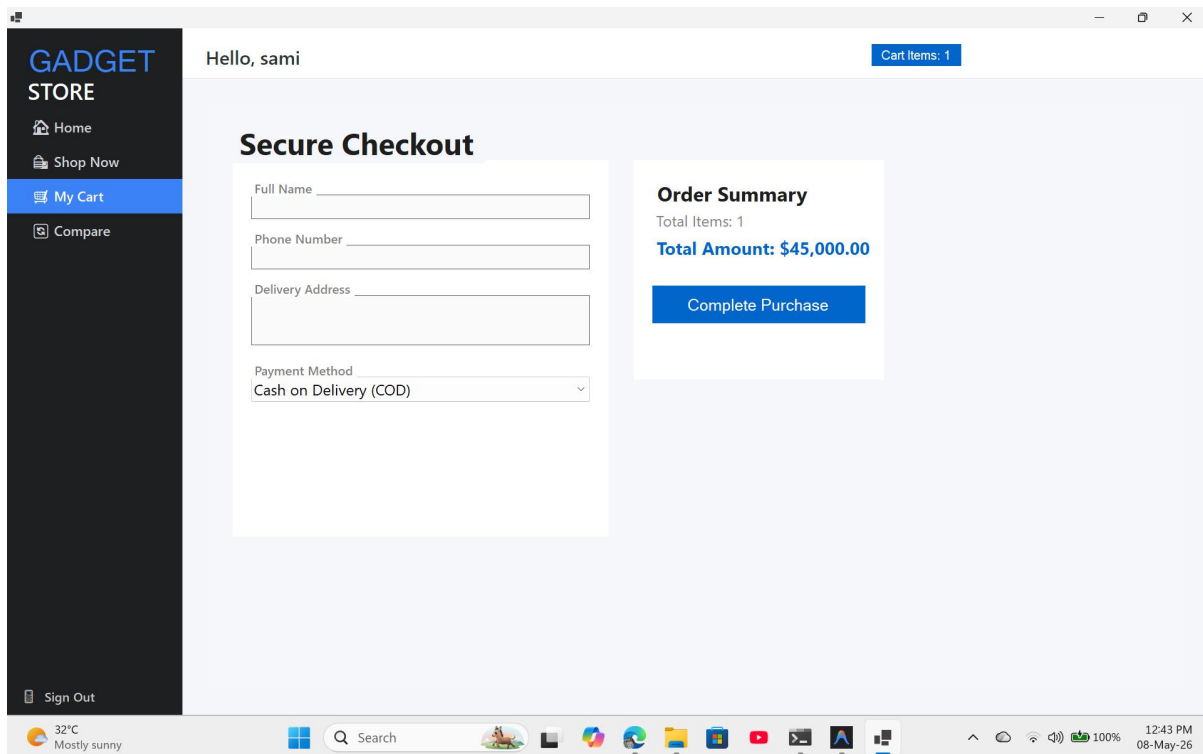
Confirm Password

☐ I agree to the Terms of Service and Privacy Policy

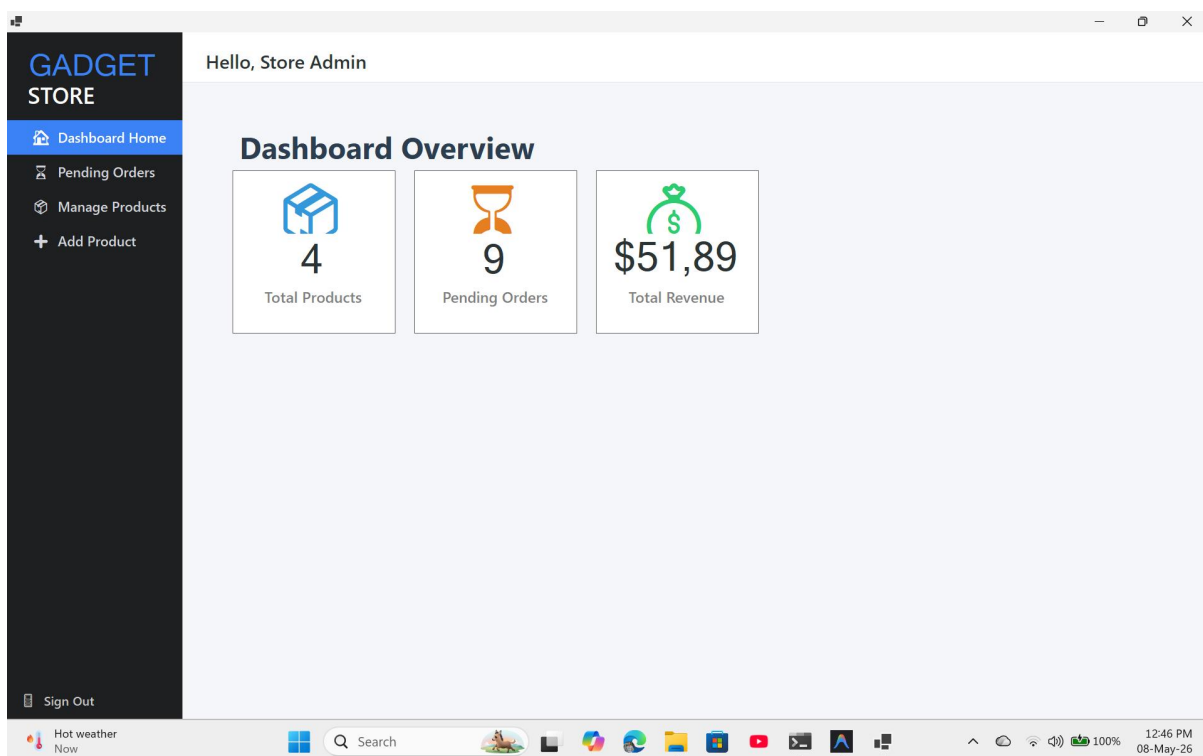
Sign Up

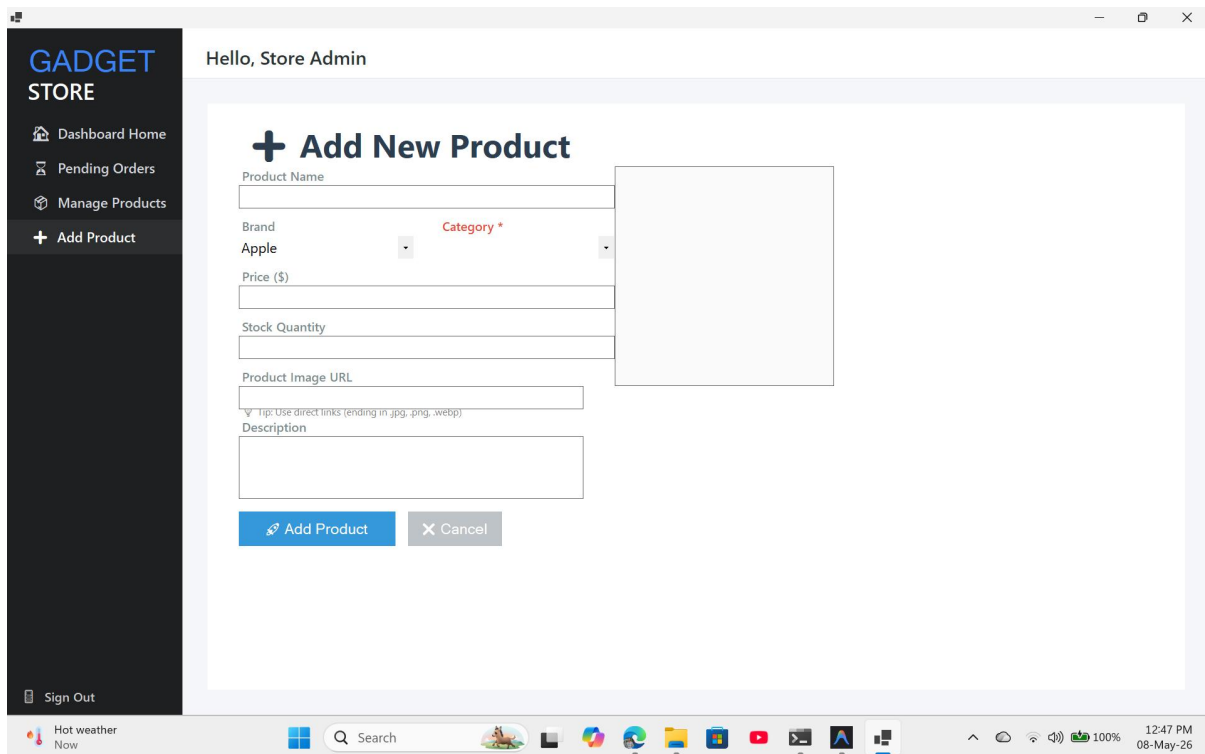




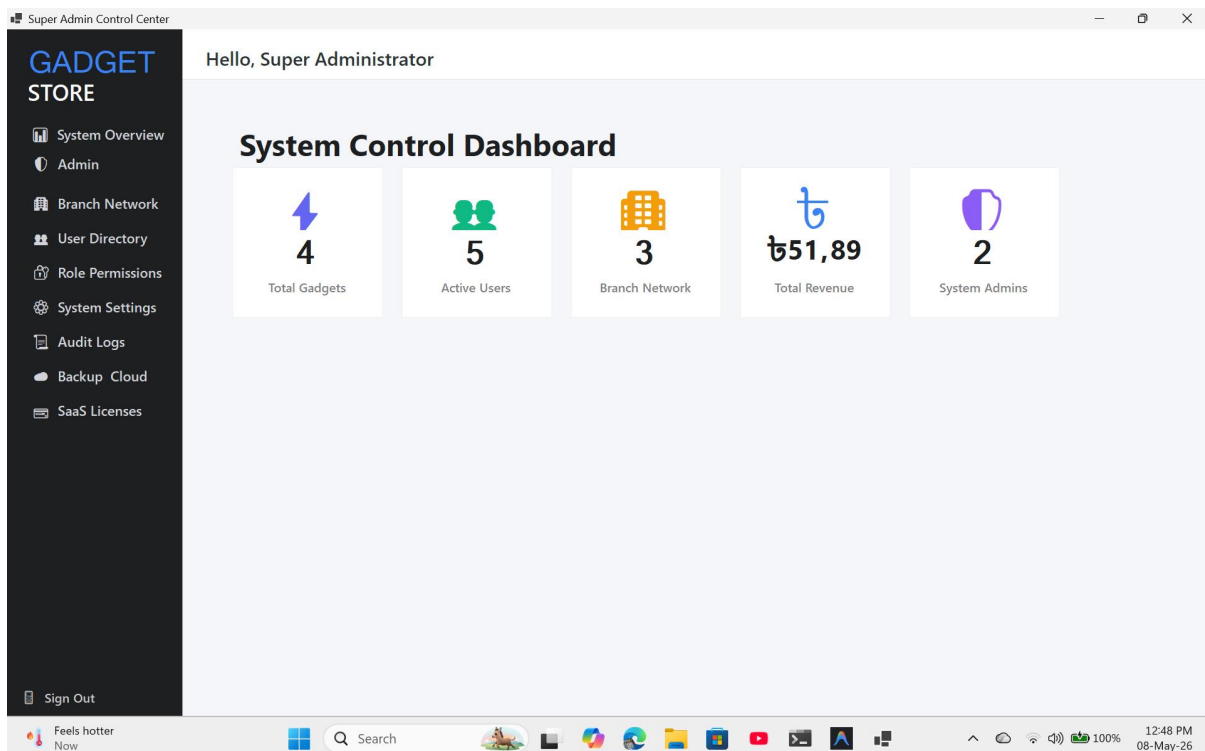


Admin Dashboards





SUPER ADMIN DASHBOARDS:



Super Admin Control Center

GADGET STORE

System Overview

Admin

Branch Network

User Directory

Role Permissions

System Settings

Audit Logs

Backup Cloud

SaaS Licenses

Sign Out

Hello, Super Administrator

	UserID	Name	Email	Role	IsLocked	CreatedAt
>	admin	Store Admin		Admin	☐	05-May-26 4:05 PM
	R1234	MD Ragib	ragibmc@gmail.com	Admin	☐	

+ Create Admin

Edit Admin

Unlock

Suspend

High UV Now

Search

12:49 PM 08-May-26

Super Admin Control Center

GADGET STORE

System Overview

Admin

Branch Network

User Directory

Role Permissions

System Settings

Audit Logs

Backup Cloud

SaaS Licenses

Sign Out

Hello, Super Administrator

	Id	Name	Location	ManagerId	IsActive	CreatedAt
>	1	Main Branch	Dhaka, Bangladesh		☒	07-May-26 6:05 PM
	2	Chittagong Branch	Chittagong, Bangladesh		☒	07-May-26 6:05 PM
	3	Rajshahi HUB	Rajshahi		☒	07-May-26 6:13 PM

Add Branch

Edit Branch

High UV Now

Search

12:50 PM 08-May-26

Super Admin Control Center

GADGET STORE

System Overview

Admin

Branch Network

User Directory

Role Permissions

System Settings

Audit Logs

Backup Cloud

SaaS Licenses

Sign Out

Hello, Super Administrator

	UserID	Name	Email	Role	IsActive
>	sami	sami	sifat	Customer	☒
	USER_5590	Sami	soms	Customer	☒
	USER_5565	SAFA	safauh1233	Customer	☒
	USER_6167	SAFA	gj	Customer	☐
	USER_1487F	Nahid	zamannahidduibhai@gmail.com	Customer	☒

Activate / Suspend Customer

High UV
Now

Search

12:50 PM
08-May-26

Super Admin Control Center

GADGET STORE

System Overview

Admin

Branch Network

User Directory

Role Permissions

System Settings

Audit Logs

Backup Cloud

SaaS Licenses

Sign Out

Hello, Super Administrator

Role-Based Access Control Management

☒ Allow Inventory Management

☒ Allow Billing System

☒ Allow Report Downloads

☒ Allow User Administration

Update Global Permissions

33°C
Mostly sunny

Search

12:50 PM
08-May-26

